Experiment 9

Aim: To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.

Theory:
Docker is a popular platform that enables developers to build, package, and deploy applications as lightweight, portable, and self-sufficient containers. These containers encapsulate all the necessary dependencies and libraries required for an application to run, ensuring consistency across different environments. Here is a theoretical overview of

Docker:
Containerization:
Docker utilizes containerization technology to create isolated environments for applications. Containers are lightweight, standalone, and executable packages that include everything needed to run an application, such as code, runtime, system tools, libraries, and settings. This isolation ensures that applications run consistently across different environments, from development to production.

Docker Engine:
At the core of Docker is the Docker Engine, which is responsible for building, running, and managing containers. It consists of the Docker daemon, which manages containers, images, networks, and volumes, and the Docker client, which allows users to interact with the daemon through the Docker API.

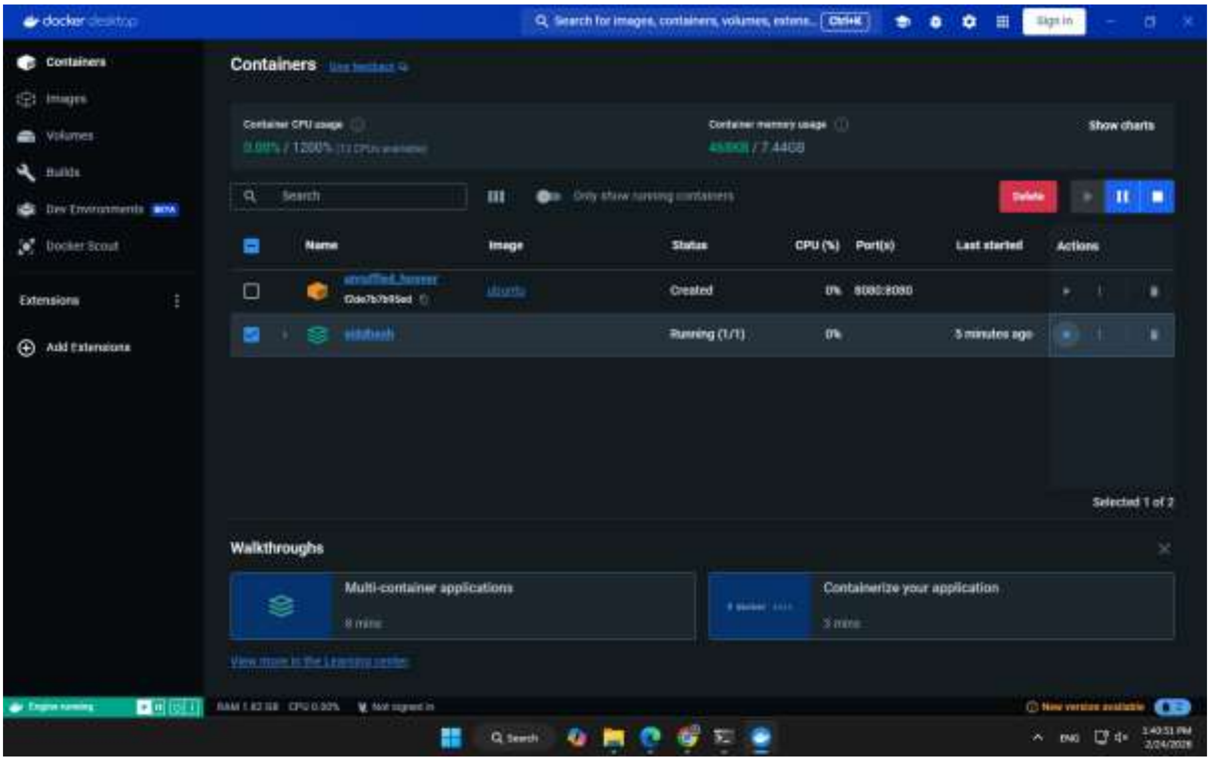
Docker Images:
Docker images are read-only templates used to create containers. They contain the application code, runtime, libraries, dependencies, and other files needed to run the application. Images are built using Dockerfiles, which are text files that define the steps needed to create the image.

Docker Containers:
Containers are instances of Docker images that are running as isolated processes on a host machine. They are lightweight, portable, and can be easily started, stopped, moved, and deleted. Containers provide a consistent environment for applications to run, regardless of the underlying infrastructure.

Benefits of Docker:

Portability: Docker containers can run on any platform that supports Docker, making it easy to deploy applications across different environments.
Efficiency: Containers share the host OS kernel, reducing overhead and improving resource utilization.

Isolation: Containers provide a level of isolation that helps prevent conflicts between applications and dependencies.
Scalability: Docker enables easy scaling of applications by quickly spinning up additional containers.
Consistency: Docker ensures that applications run the same way in development, testing, and production environments.



```
C:\Users\Lenovo>docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Log in to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  builder     Manage builds
  buildx*     Docker Buildx (Docker Inc., v0.12.1-desktop.4)
  compose*    Docker Compose (Docker Inc., v2.24.6-desktop.1)
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container. (Docker Inc., 0.0.24)
  dev*        Docker Dev Environments (Docker Inc., v0.1.0)
  extension*  Manages Docker extensions (Docker Inc., v0.2.22)
  feedback*   Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
  image       Manage images
  init*       Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
  manifest    Manage Docker image manifests and manifest lists
  network     Manage networks
  plugin      Manage plugins
  sbom*       View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
  scout*      Docker Scout (Docker Inc., v1.5.0)
  system      Manage Docker
  trust       Manage trust on Docker images
  volume      Manage volumes
```

```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker search ubuntu
NAME                             DESCRIPTION                                     STARS     OFFICIAL
ubuntu                           Ubuntu is a Debian-based Linux operating sys…   17789     [OK]
ubuntu/squid                     Squid is a caching proxy for the Web. Long-t…   124
ubuntu/nginx                     Nginx, a high-performance reverse proxy & we…   139
ubuntu/cortex                    Cortex provides storage for Prometheus. Long…   4
ubuntu/bind9                     BIND 9 is a very flexible, full-featured DNS…   117
ubuntu/kafka                     Apache Kafka, a distributed event streaming …   60
ubuntu/zookeeper                 ZooKeeper maintains configuration informatio…   14
ubuntu/apache2                   Apache, a secure & extensible open-source HT…   102
ubuntu/prometheus                Prometheus is a systems and service monitori…   78
ubuntu/mysql                     MySQL open source fast, stable, multi-thread…   72
ubuntu/dotnet-aspnet             Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/jre                       Chiseled Java runtime based on Ubuntu. Long-…   24
ubuntu/postgres                  PostgreSQL is an open source object-relation…   42
ubuntu/python                    A chiselled Ubuntu rock with the Python runt…   31
ubuntu/redis                     Redis, an open source key-value store. Long-…   24
ubuntu/dotnet-runtime            Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps               Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                   Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                 Memcached, in-memory keyvalue store for smal…   6
ubuntu/prometheus-alertmanager   Alertmanager handles client alerts from Prom…   10
ubuntu/cassandra                 Cassandra, an open source NoSQL distributed …   3
ubuntu/mlflow                    MLFlow: for managing the machine learning li…   6
ubuntu/loki                      Grafana Loki, a log aggregation system like …   2
ubuntu/mimir                     Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/opentelemetry-collector   This is a rock for the OpenTelemetry Collect…   0
```

```
C:\Users\Lenovo>docker search --filter=stars=1000 ubuntu
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17789     [OK]
```

```
C:\Users\Lenovo>docker search mysql
NAME                  DESCRIPTION                                     STARS     OFFICIAL
mysql                 MySQL is a widely used, open-source relation…   10070     [OK]
bitnami/mysql         Bitnami Secure Image for mysql                  147
circleci/mysql        MySQL is a widely used, open-source relation…   32
bitnamicharts/mysql   Bitnami Helm chart for MySQL                    0
cimg/mysql                                                            3
ubuntu/mysql          MySQL open source fast, stable, multi-thread…   72
google/mysql          MySQL server for Google Compute Engine          26
linuxserver/mysql     A Mysql container, brought to you by LinuxSe…   48
elestio/mysql         Mysql, verified and packaged by Elestio         3
alpine/mysql          mysql client                                    7
docksal/mysql         MySQL service images for Docksal - https://d…   0
eclipse/mysql         Mysql 5.7, curl, rsync                          3
ilios/mysql           Mysql configured for running Ilios              1
datajoint/mysql       MySQL image pre-configured to work smoothly …   2
ddev/mysql            ARM64 base images for ddev-dbserver-mysql-8.…   2
mirantis/mysql                                                        0
mysql/mysql-server    Optimized MySQL Server Docker images. Create…   1033
corpusops/mysql       https://github.com/corpusops/docker-images/     0
cleanstart/mysql      Secure by Design, Built for Speed, Hardened …   0
vulhub/mysql                                                          1
cbioportal/mysql      This repository hosts MySQL database images …   1
vitess/mysql          Lightweight image to run MySQL with Vitess      1
mysql/mysql-router    MySQL Router provides transparent routing be…   28
mysql/mysql-cluster   Experimental MySQL Cluster Docker images. Cr…   100
nasqueron/mysql                                                       1
```

```
C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker images
REPOSITORY                       TAG                                        IMAGE ID       CREATED         SIZE
ubuntu                           latest                                     bbdabce66f1b   13 days ago     78.1MB
ubuntu                           <none>                                     802541663949   6 months ago    78.1MB
ubuntu                           <none>                                     ca2b0f26964c   24 months ago   77.9MB
redis                            <none>                                     170a1e90f843   2 years ago     138MB
docker/dev-environments-default  stable-1                                   7eb05f44c229   3 years ago     610MB
crazymax/linguist                7.20.0                                     dc5f9ae3be6a   3 years ago     72.2MB
docker/desktop-git-helper        5a4fca126aadcd3f6cc3a011aa991de982ae7000   efe2d67c403b   4 years ago     44.2MB

C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit

C:\Users\Lenovo>docker run -it ubuntu
root@2ddd2c7718ec:/# admin
bash: admin: command not found
root@2ddd2c7718ec:/# ^C
root@2ddd2c7718ec:/# exit
exit

C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d828aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a398514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
9f10642c64c6c3743dc235fe26caf0a43a9876c261d5b29b28a34b046922393d
```

```
C:\Users\Lenovo>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658638c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b38d8ef2c592e63e4e8a9a52658a
Status: Downloaded newer image for mysql:latest
af5bbf473e88d0b9a7264168926d9712c94a6d94497687fed89179d31e778ccd

C:\Users\Lenovo>docker run -d ubuntu
17ae4f0c3240c5b5ba81df3d23be8a1eef290d48680f89b473355676855140b7a

C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE                                     COMMAND                CREATED              STATUS                 PORTS     NAMES
9f10642c64c6   httpd                                     "httpd-foreground"     About a minute ago   Up About a minute      80/tcp    wonderful_babbage
a388ed3e2ae7   docker/dev-environments-default:stable-1  "sleep infinity"       23 months ago        Up 12 minutes                    siddhesh-app-1
```